

PrimeFocus

AUGUST 2025



WHEN

August 15, 2025

Doors open at 7:00pm

Meeting starts at 7:30pm

Lecture starts at 8:00pm

WHERE

Unitarian Church

1893 North Vasco Road

Livermore, CA 945571

and via Zoom

TVS QR CODE



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THIS MONTH'S SPEAKER

The Secrets of Long Focal Length Imaging John Hayes, Ph.D.

Long focal length deep sky imaging presents special challenges and rewards. It's possible to produce stunning images of many galaxies and smaller nebula; however, there are many challenges that must be addressed to achieve good results. We'll review mechanical stability requirements, precision optical alignment, guiding methods, dealing with thermal issues, selecting the best camera and how seeing affects results. The author will pass along a number of recommendations, tips and tricks learned from imaging with a highly modified C14 EdgeHD, a 20" CDK, and an ASA600--all working at an EFL of around 4 m.

We'll review:

1. Optimizing your optical system to achieve the best possible image quality.
2. Picking the best camera and sampling to get the best SNR and highest resolution.
3. Understanding the seeing blur disk and its impact on system design and performance.
4. Configuring the mechanics to maximize stability and minimize vibration in a world made of rubber.
5. Guiding and focusing strategies to maximize image quality.
6. Tips and tricks to optimize system performance--what's important and what's not. Along the way, we'll look at some common mistakes and how to avoid them.
7. Thermal sensitivity and how it constrains maximum exposure time.

THE SECRETS OF LONG FOCAL LENGTH IMAGING



John Hayes, Ph.D.

Adjunct Research Professor
College of Optical Sciences
University of Arizona

2025



A blurry eyed portrait after a week of sleepless nights working on my telescope in Chile!
Credit: John Hayes

John Hayes is a retired optical engineer, entrepreneur, and professor with a B.S. in physics and astronomy, a M.S. and Ph.D. in optics from the University of Arizona. His interest in telescopes and astronomy started in elementary school, which led to a career in the field of optical engineering. He has authored 30 professional papers, articles, book chapters, and presentations in the field of optical testing and holds 14 patents in the fields of optics and atomic force microscopy. In retirement, Dr. Hayes has returned to the hobby of astro-imaging, which he does with an ASA600 on a L-600 mount and a GTX130 refractor on a Mach2 mount--both at Obstech in Chile. He has made numerous presentations on the Astro-Imaging Channel, AIC, NEIAC and to numerous other clubs, universities and organizations. He has been recognized with 23 IOTD awards and 2 NASA APODs. He is also an active pilot and flight instructor with jet type ratings and a former regular contributor to Plane and Pilot magazine. As an Adjunct Research Professor of Optics, he enjoys using his technical background to help amateurs explore and advance the state of the art of astro-imaging.

NEWS AND NOTES

Meeting Dates

Club Meeting	Board Meeting	PrimeFocus Deadline
August 15	August 18	August 5
September 19	September 22	September 6
October 17	October 20	October 4

Money Matters

As of the last Treasurer's Report on 7/21/25, our club's account balance is \$47,357.42, this includes \$11,105.46 in the H2O Rebuild fund.

TVS Welcomes New Members

TVS welcomes new members Santosh Murthy, Munish Kapoor, Jake Harm, and Raven Merrill-LaRue. Please say hello and chat with them during our meetings.

2025 TVS Club Star Party Schedule

Save the dates for the 2025 Club Star Parties.

Del Valle star parties are also public outreach events. They are jointly hosted with the EBRPD and held at the Arroyo Staging Area (Coords: 37.6196638, -121.7528899). The public is invited for the first 1.5-2 hours, while club members can stay the remainder of the night.

August 30: Del Valle Star Party, Arroyo Road Staging Area., Livermore. 7:40pm-10:00pm.

Tesla Vintners star parties are open to only club members and their guests. These star parties end at midnight, but participants can leave earlier, should they wish.

September 13: Tesla Vintner's Star Party, 5143 Tesla Rd., Livermore. Set-up at 6:30pm, Observing 7:00pm-11:30pm.

H2O Open House star parties are open to only club members and their guests. The open house ends at midnight, and all participants are encouraged to stay the duration. The drive to H2O takes about 1 hour, and the caravan leaves promptly from the corner of Mines and Tesla Rds. No gas stations are available on the route, so be prepared. Admission is \$3/car-bring exact change. H2O is a primitive site with two porta-potties. Bring water, food, and warm clothing, as needed. Red flashlights are to be used so observers can preserve their night vision.

August 16: H2O Open House, 5:00pm caravan to H2O PROMPTLY leaves the corner of Mines and Tesla Rds., Livermore. Observing until 11:30pm. Meeting times are tentative.

September 30: Public Star Party at Sunol Regional Wilderness, 1895 Geary Road, Sunol ([Google Map](#)). Set-up 6:45pm, Observing from 7:30-9:30pm.

CALENDAR OF EVENTS



August 22, 23, 29,30, September 5, 6, 12,13, 19, 20, 7:30-10:30pm

What	Free Telescope Viewing
Who	Chabot Staff
Where	Chabot Space and Science Center, 10000 Skyline Blvd. Oakland, CA 94619
Cost	Free

Join Chabot astronomers on the Observatory Deck for a free telescope viewing! Weather permitting, this is a chance to explore stars, planets and more through Chabot's historic telescopes. Chabot's three large historic telescopes offer a unique way to experience the awe and wonder of the Universe. Three observatory domes house the Center's 8-inch (Leah, 1883) and 20-inch (Rachel, 1916) refracting telescopes, along with a 36-inch reflecting telescope (Nellie, 2003).

Are the skies clear for viewing tonight? Viewing can be impacted by rain, clouds, humidity and other weather conditions. Conditions can be unique to Chabot because of its unique location in Joaquin Miller Park. Before your visit, check out the Weather Station to see the current conditions at Chabot.

For more information, see <https://chabotspace.org/events/events-listing/>
<https://eastbayastro.org/chabot-telescope-status/>

September 8, 7:30 PM

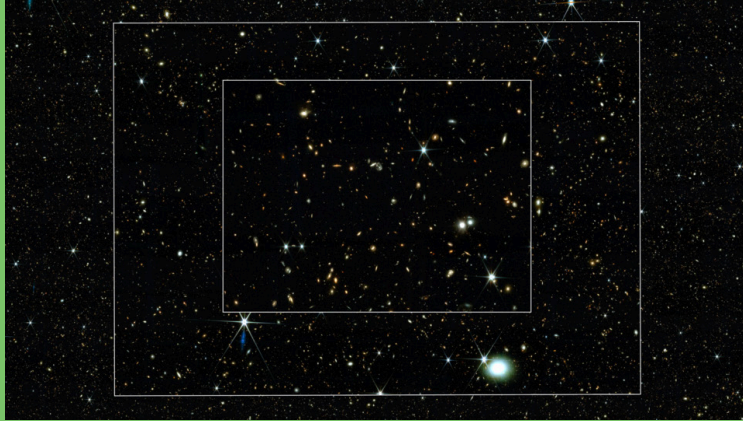
What	Formed Too Fast? The Biggest Galaxies at Cosmic Dawn
Who	California Academy of Sciences
Where	Morrison Planetarium; 55 Music Concourse Drive, San Francisco, CA 94118
Cost	Public: \$15; Members and seniors: \$12

Featuring Dr. Caitlin Casey, UC Santa Barbara

The incredible capabilities of the James Webb Space Telescope have allowed us to look back nearly to the very beginning of time, enabling us to observe some of the first galaxies that formed after the Big Bang. Before the launch of JWST, astronomers believed that the first few hundred million years—when the

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Universe was only a small fraction of its current age—were relatively quiet. To our surprise, we have discovered many more bright galaxies than previously expected. These bright galaxies challenge our understanding of how the first galaxies and black holes formed. In this talk, Dr. Casey will guide us through the journey of discovery in the universe's earliest moments and address some of the key questions raised by JWST's groundbreaking findings.



This image shows a progressive zoom-in on JWST's COSMOS-Web deep field, which is 200 times larger than the Hubble Deep Field and contains nearly a million galaxies from the early universe to today. Its size helps astronomers locate distant objects and learn about the environments where they exist on the largest scales known.

Caitlin Casey is an observational astronomer and Professor of Physics at the University of California, Santa Barbara. Before moving to UCSB in 2025, she was an astronomy professor at the University of Texas at Austin since 2015, and previously both a NASA Hubble postdoctoral fellow at the University of Hawai'i and McCue Fellow of Cosmology at the University of California, Irvine. Casey received her PhD in Astronomy from the University of Cambridge in 2010 funded as a Gates Cambridge Scholar by the Gates Foundation; she received her B.S. in Physics, Astronomy, and Applied Mathematics from the University of Arizona in 2007. She was awarded the 2018 Newton Lacy Pierce Prize for her pioneering contributions to observational astronomy before age 36. She has expertise in the most massive galaxies formed through cosmic time and uses an array of telescopes around the world and in space to answer fundamental questions about the growth of the first galaxies. Casey leads one of the largest observational projects on JWST called COSMOS-Web, which is a deep field about 200 times larger than the Hubble Deep Field.



For more information, see: <https://www.calacademy.org/events/benjamin-dean-astronomy-lectures/formed-too-fast-the-biggest-galaxies-at-cosmic-dawn>



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Milky Way rising over Granite Mountain, By Roland Albers.



The Squid Nebula, By Ashish Joshi.
For full size image, see <https://www.astrobin.com/8uqlm1/B/>



M27 The Dumbbell Nebula, By Ishan Gole.



The Eagle Nebula, By Ishan Gole.



The Swan Nebula, By Ishan Gole.



M31, M32, & M110, By Chris Archer.

WHATS UP

Adapted from Sky and Telescope

All times are Pacific Standard Time

August 2025

20	Wed	In the morning facing East Northeast, see the Moon about 4° upper left of Venus and Pollux is about 2½° to the left of the Moon with Jupiter to the right.
21	Thu	At dawn the Mercury is 4½° to the lower right of the Moon
23	Sat	New Moon
26	Tue	At dusk looking west, crescent Moon is less than 6° lower left of Mars.
30	Sat	At dusk moon is just 4½° lower right of Antares
31	Sun	Moon is at first quarter

September 2025

7	Sun	Full Moon
8	Mon	At dusk, Saturn, just 5½° to the right, rises with the Moon.
12	Fri	Moon follows the Pleiades by 3° as they rise above the east-northeast horizon.
13	Sat	Looking low in the west southwest, Spica and Mars are just 2° apart.
14	Sun	Moon at last quarter

OFFICERS AND VOLUNTEER POSITIONS

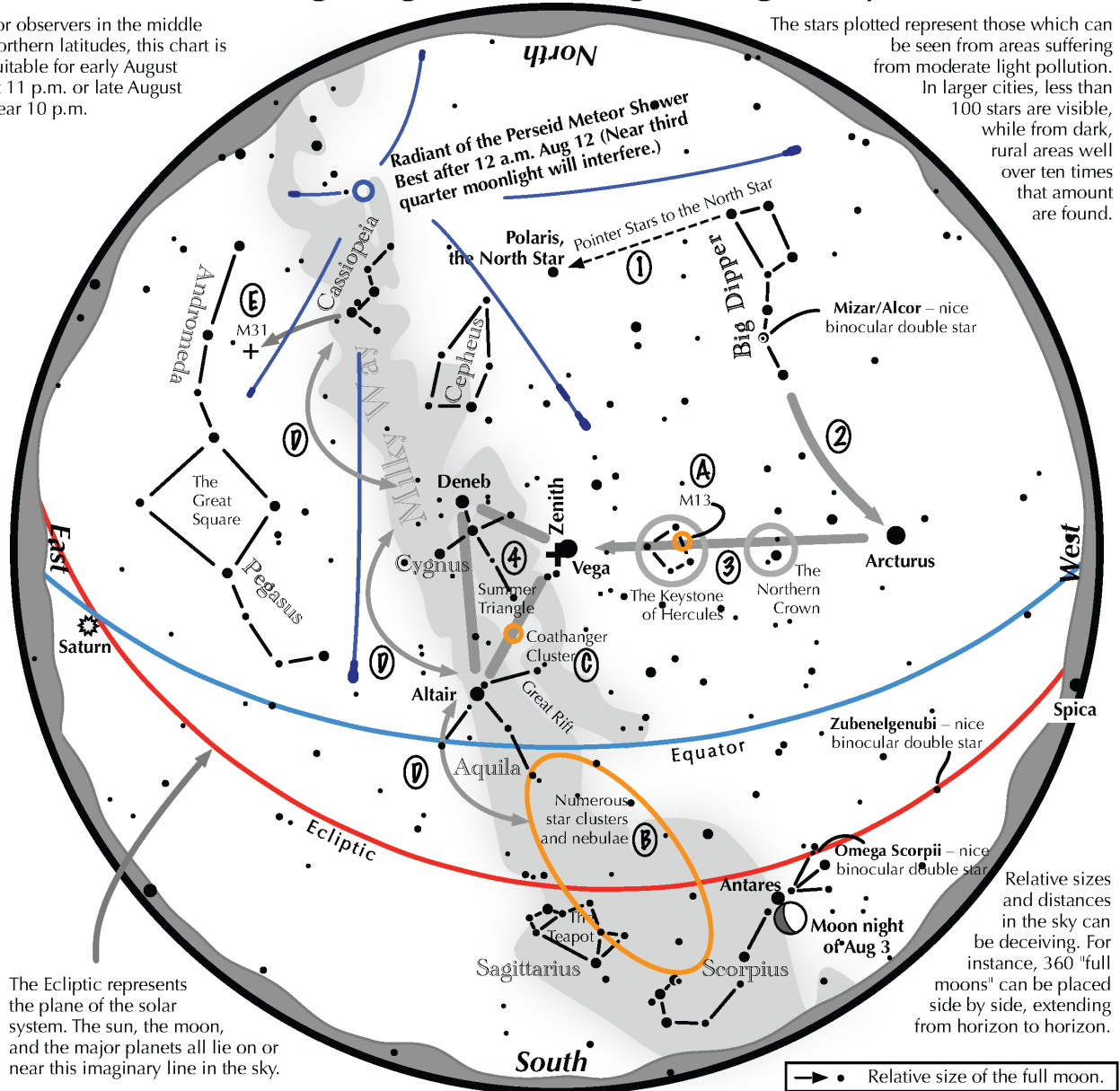
Officers	Volunteer Positions	Newsletter	Potluck Coordinator
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Secretary Dave Lackey secretary@trivalleystargazers.org	Historian OPEN historian@trivalleystargazers.org	H2O Observatory Director / Rebuild Chairman Chuck Grant H20@trivalleystargazers.org	Refreshment Coordinator OPEN
Past President Ron Kane past_president@trivalleystargazers.org	Librarian Ron Kane librarian@trivalleystargazers.org	Observing Program Coordinator Ron Kane awards@trivalleystargazers.org	Web and Email www.trivalleystargazers.org info@trivalleystargazers.org
	Loaner Scope Manager Ron Kane telescopes@trivalleystargazers.org	Outreach Coordinator Johnathan Bailey outreach@trivalleystargazers.org	TVS E-Group To Join the TVS E-Group just send an email to TVS at info@trivalleystargazers.org asking to join the group. Make sure you specify the email address you want to use to read and post to the group.

NAVIGATING THE NIGHT SKY FOR AUGUST

Navigating the mid August Night Sky

For observers in the middle northern latitudes, this chart is suitable for early August at 11 p.m. or late August near 10 p.m.

The stars plotted represent those which can be seen from areas suffering from moderate light pollution. In larger cities, less than 100 stars are visible, while from dark, rural areas well over ten times that amount are found.



Navigating the mid August night sky: Simply start with what you know or with what you can easily find.

- 1 Extend a line north from the two stars at the tip of the Big Dipper's bowl. It passes by Polaris, the North Star.
- 2 Follow the arc of the Dipper's handle. It intersects Arcturus, the brightest star in the June evening sky.
- 3 To the northeast of Arcturus shines another star of the same brightness, Vega. Draw a line from Arcturus to Vega. It first meets "The Northern Crown," then the "Keystone of Hercules." A dark sky is needed to see these two dim stellar configurations.
- 4 High in the East lies the summer triangle stars of Vega, Altair, and Deneb.

Binocular Highlights

- A:** On the western side of the Keystone glows the Great Hercules Cluster.
- B:** Between the bright stars Antares and Altair, hides an area containing many star clusters and nebulae.
- C:** 40% of the way between Altair and Vega, twinkles the "Coathanger," a group of stars outlining a coathanger.
- D:** Sweep along the Milky Way for an astounding number of faint glows and dark bays, including the Great Rift.
- E:** The three westernmost stars of Cassiopeia's "W" point south to M31, the Andromeda Galaxy, a "fuzzy" oval.

Astronomical League www.astroleague.org/outreach; duplication is allowed and encouraged for all free distribution.



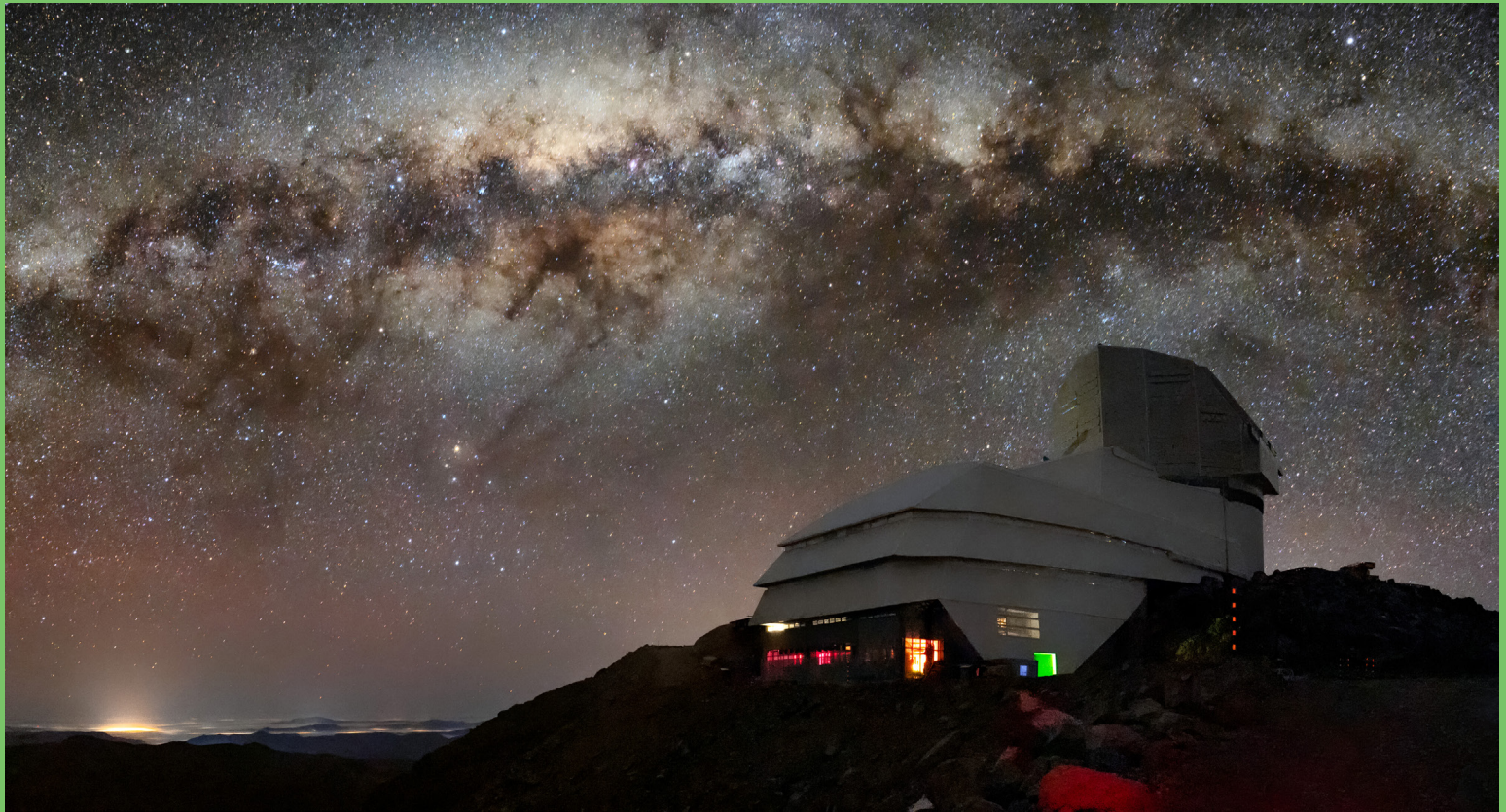
NASA NIGHT SKY NOTES

The Great Rift

By **Dave Prosper**, Updated by **Kat Troche**

Summer skies bring glorious views of our own Milky Way galaxy to observers blessed with dark skies. For many city dwellers, their first sight of the Milky Way comes during trips to rural areas - so if you are traveling away from city lights, do yourself a favor and look up!

To observe the Milky Way, you need clear, dark skies and enough time to adapt your eyes to the dark. Photos of the Milky Way are breathtaking, but they usually show far more detail and color than the human eye can see – that's the beauty and quietly deceptive nature of long exposure photography. For Northern Hemisphere observers, the most prominent portion of the Milky Way rises in the southeast as marked by the constellations Scorpius and Sagittarius. Take note that, even in dark skies, the Milky Way isn't easily visible until it rises a bit above the horizon, and the thick, turbulent air obscures the view. The Milky Way is huge, but it is also rather faint, and our eyes need time to truly adjust to the dark and see it in any detail. Avoid bright lights as they will ruin your night vision. It's best to attempt to view the Milky Way when the Moon is at a new or crescent phase; a full Moon will wash out any potential views.



The Vera C. Rubin Observatory, located at Cerro Pachón, Chile, under the Milky Way. The bright halo of gas and stars on the left side of the image highlights the very center of the Milky Way galaxy. The dark path that cuts through this center is known as the Great Rift, because it gives the appearance that the Milky Way has been split in half. Image Credit: [RubinObs/NOIRLab/SLAC/NSF/DOE/AURA/B. Quint](https://www.noirlab.si.edu/observers/rubin)

Keeping your eyes dark-adapted is especially important if you want to not only see the haze of the Milky Way, but also the dark lane cutting into that haze, stretching from the Summer Triangle to Sagittarius. This dark detail is known as the Great Rift, and is seen more readily in very dark skies, especially dark, dry skies found in high desert regions. What exactly is the Great Rift? You are looking at massive clouds of galactic dust lying between Earth and the interior of the Milky Way.

Other “dark nebulae” of cosmic clouds pepper the Milky Way, including the famed Coalsack, found in the Southern Hemisphere constellation of Crux. Many cultures celebrate these dark clouds in their traditional stories along with the constellations and the

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The Great Rift, continued

Milky Way. One such story tells of a Yacana the Llama, and her baby, wandering along a river that crossed the sky – the Milky Way. The bright stars Alpha and Beta Centauri serve as the llama's eyes, with the dark sections representing the bodies of mother and baby, with the baby below the mother, nursing.



In the activity, "Our Place In Our Galaxy", if the Milky Way were shrunk down to the size of North America, our solar system would be about the size of a quarter. At that scale, Polaris - which is about 433 light years distant from us - would be 11 miles away. Image Credit: [Astronomical Society of the Pacific](#)

Where exactly is our solar system within the Milky Way? Is there a way to get a sense of scale? The "Our Place in Our Galaxy" activity can help you do just that, with only birdseed, a coin, and your imagination. You can also discover the amazing science NASA is doing to understand our galaxy – and our place in it - in the Galaxies section of NASA's Universe page.

Originally posted by Dave Prosper: June 2021

Last Updated by KatTroche: July 2025



This article is distributed by NASA's Night Sky Network (NSN).

The NSN program supports astronomy clubs across the USA dedicated to astronomy outreach. Visit nightsky.jpl.nasa.gov to find local clubs, events, and more!



Tri-Valley Stargazers
P.O. Box 2476
Livermore, CA 94551
www.trivalleystargazers.org

Tri-Valley Stargazers Membership Application

Contact information:

Name: _____ Phone: _____

Street Address: _____

City, State, Zip: _____

Email Address: _____

Status (select one): _____ New member _____ Renewing or returning member

Membership category (select one): Membership term is for one calendar year, January through December.

_____ Student member (\$10). Must be a full-time high-school or college student.

_____ Regular member (\$30).

Hidden Hill Observatory Access (optional): Must be 18 or older.

_____ One-time key deposit (\$20). This is a refundable deposit for a key to H2O. New key holders must first hear an orientation lecture and sign a usage agreement form before using the observing site.

_____ Annual access fee (\$10). You must also be a key holder to access the site.

Donation (optional):

_____ Tax-deductible contribution to Tri-Valley Stargazers

Total enclosed: \$ _____

Member agrees to hold Tri-Valley Stargazers, and any cooperating organizations or landowners, harmless from all claims of liability for any injury or loss sustained at a TVS function. TVS will not share information with anyone except as detailed in our Privacy Policy (<http://www.trivalleystargazers.org/privacy.shtml>).

Mail this completed form along with a check to: Tri-Valley Stargazers, P.O. Box 2476, Livermore, CA 94551.